



■ CS177 Bioinformatics: Software Development and Applications

Lecture 20: Systems Biology

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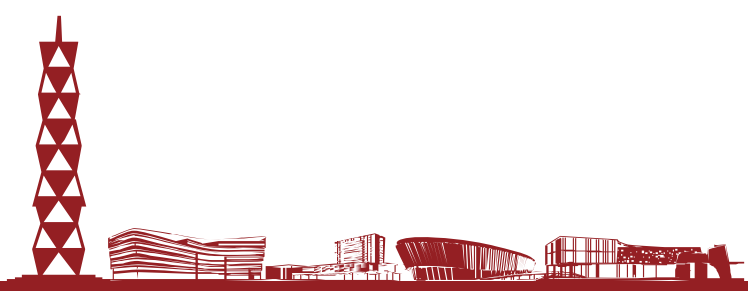
Spring, 2025



Learning objectives



- At the end of this lecture, you should be able to:
 - Explain what is systems biology
 - Give a list of different types of biological networks
 - Analyze the topological characteristics and structures of biological networks
 - Describe the basics of deterministic modeling, e.g. the Michaelis-Menten model
 - Describe a case of using LLM to infer gene regulatory networks (GRNs), i.e. LLM4GRN





Principles of Systems Biology



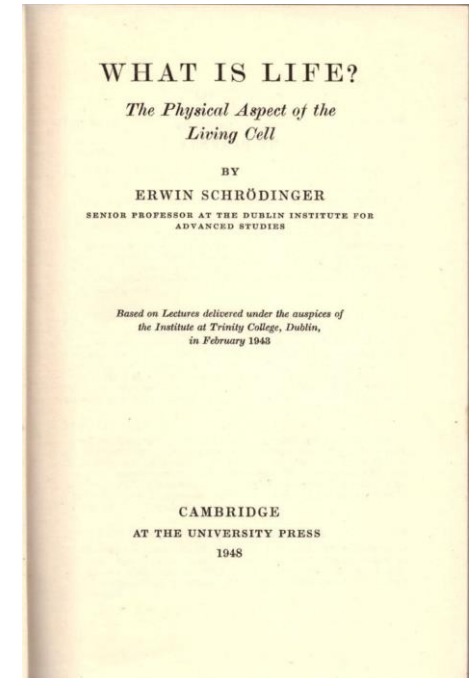


What is Systems Biology?

- **Systems biology** is concerned with the study of biological functions and mechanisms, underpinning inter- and intra-cellular dynamical networks, by means of system-oriented approaches
- *"Life is an emergent, rather than an immanent and inherent, property of matter: Although it arises from the material world, it cannot be reduced to it." (Erwin Schrödinger, in his classic book **"What is Life?"** , 1944)*
- A systems biology approach means
 - Investigating the components of cellular networks and their **interactions**
 - Applying experimental high-throughput and whole-genome techniques
 - **Integrating** computational and theoretical methods with experimental efforts



Erwin Schrödinger (1887 - 1961), Austrian theoretical physicist.



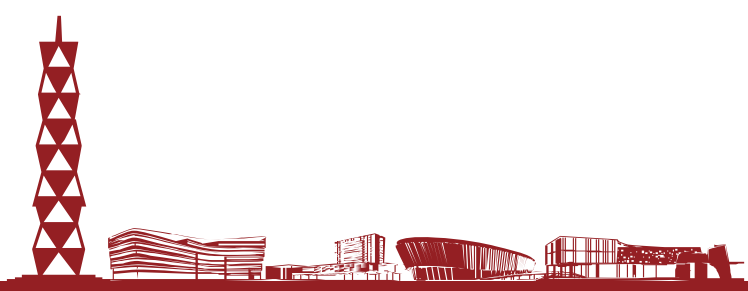


Applications of systems biology



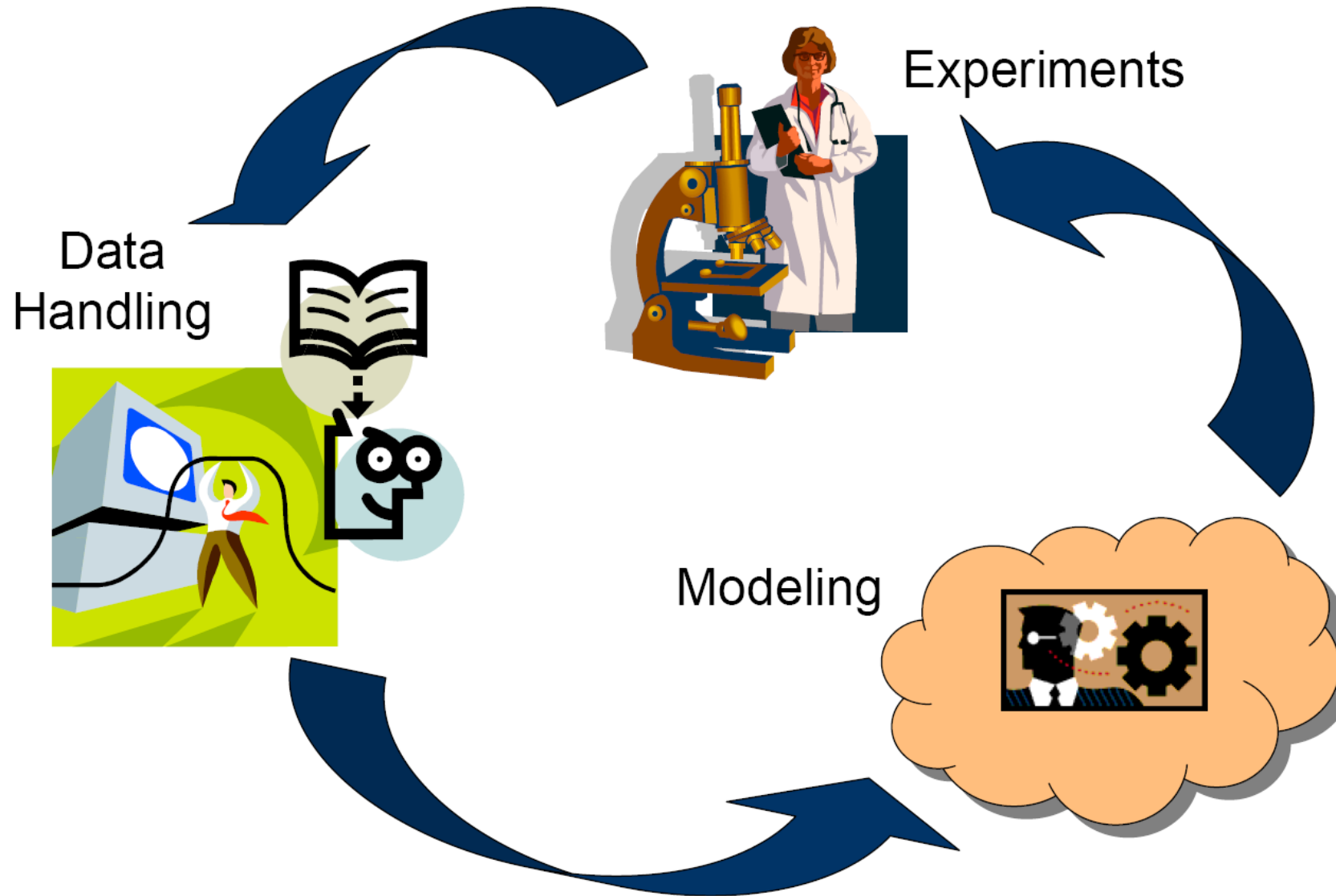
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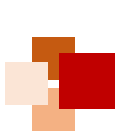
- Construct models to unveil mechanisms causing altered phenotypes to help devise novel therapies and drugs
- Predictive tools to design cells with desired properties cheaply and reliably
- Personalized and predictive medicine



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An iterative approach





What is a model?



- It depends on whom you ask ...
 - **Geneticist:** the mouse family Ts65DN serves as a model for human trisomy 21
 - **Chemist:** a **reaction network**, described by dots (for metabolites) and arrows (for reactions)
 - **Mathematician/Engineer:** the same reaction network can be modeled by a system of nonlinear ODEs (**ordinary differential equations**)
- Abstract representation of objects or processes that explains features of these objects or processes





Models are not real, but reality can be modeled



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- In almost any case, models are only rough representations of their real-world (e.g. biological) counterparts
- Nevertheless, models allow us to
 - Check the reliability of basic assumptions
 - Uncover the lack of knowledge and requirements for clarification
 - Elucidate underlying mechanisms and properties
 - Create large repositories of current knowledge, formalized in a non-ambiguous way and including quantitative data



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Overview of Biological Networks





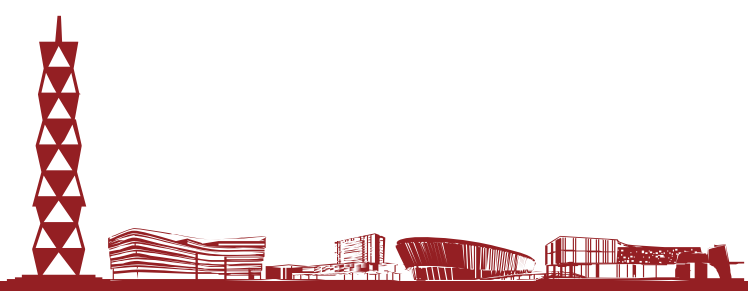
Types of biological networks



- Different kinds of biological networks can be distinguished at the molecular level
 - Gene regulatory network
 - Metabolic network
 - Signal transduction network
 - Protein-protein interaction network
- Moreover, other networks can be considered as we move to different description levels, e.g.
 - Immunological network
 - Neural network
 - Ecological network
- Here, we will focus exclusively on molecular processes that take place **within** the cell



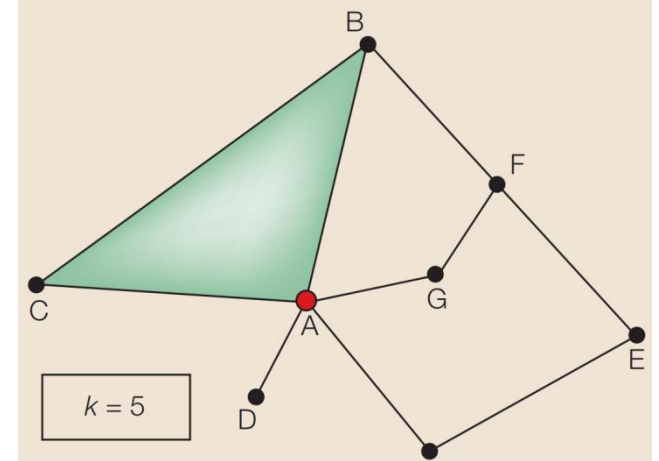
- A major challenge is to **infer** the networks of complex macromolecular interactions at the gene, metabolite and protein levels, with reasonable accuracy
- Once identified, a network model can be used to:
 - **Simulate** the biological process it represents
 - **Predict** the features of its dynamical behavior
 - **Extrapolate** cellular phenotypes



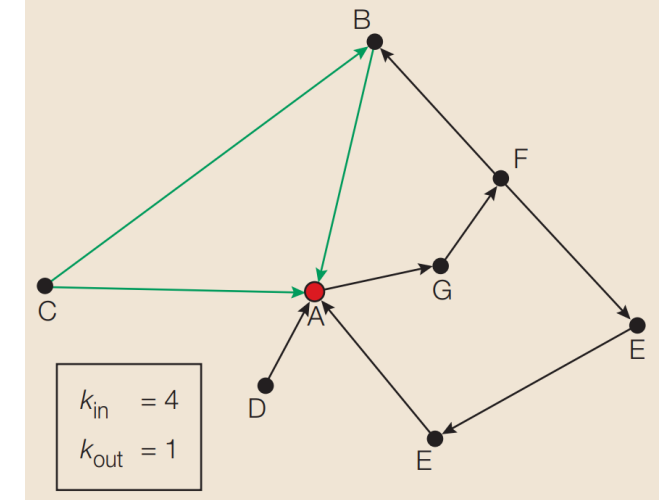
Graphs

- Graph representation is a very useful formal tool for describing and visualizing biological networks
 - An **(undirected) graph** is an ordered pair $G = (V, E)$, where V is the set of the **vertices** (or **nodes**), and E is the set of unordered pairs of distinct vertices, called **edges** (or **lines**)
 - For each edge $\{u, v\}$, the nodes u and v are said to be **adjacent**
 - If E is a set of ordered pairs, we have a **directed graph**, or **digraph**
 - In digraphs, the **in-degree**, k_{in} , (or, **out-degree**, k_{out}) of a node is the number of edges entering into (or, leaving from) that node

a Undirected network



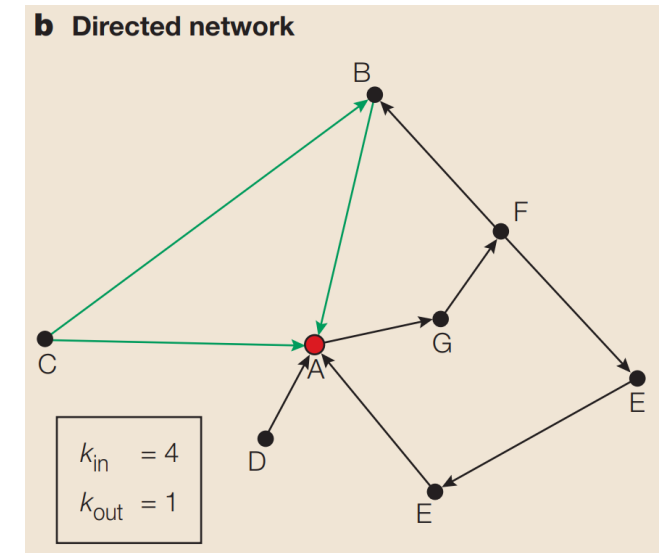
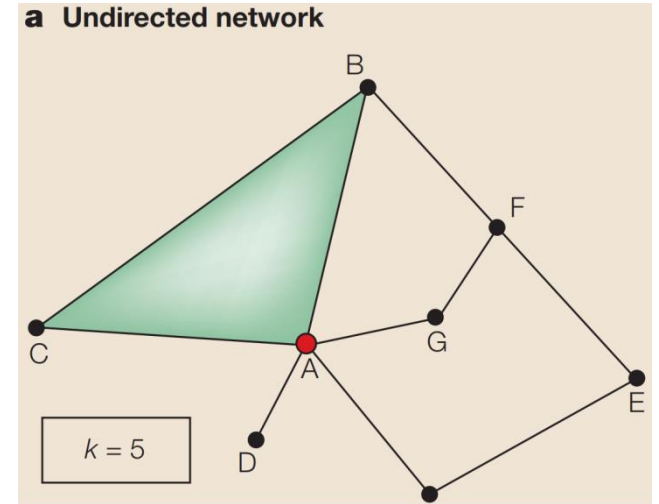
b Directed network



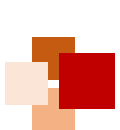
Barabasi et al. Nature Review
Genetics, 101(5), 101-113, 2004

Topological characteristics

- The **degree distribution**, $P(k)$, gives the probability that a selected node has exactly k links
- It allows us to distinguish between different classes of networks (see next slide)
- The **clustering coefficient** of a node I , C_I , measures the **aggregation** of its adjacencies (i.e. the number of triangles passing through node I)
- $C(k)$ is the average clustering coefficient of all nodes with k links



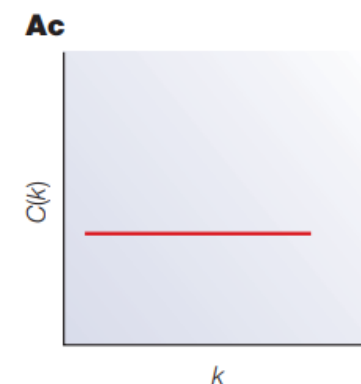
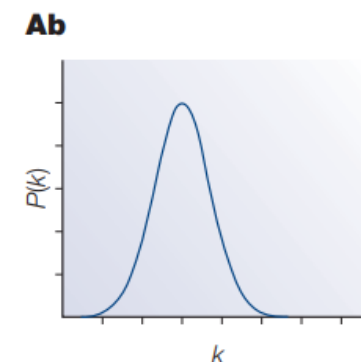
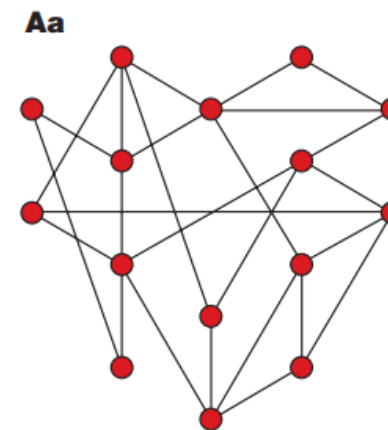
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Erdős-Rényi random networks



- The Erdős-Rényi model of a random network starts with N nodes and connects each pair of nodes with probability p
- The degree allows a **Poisson** distribution, thus many nodes have the same number of links (close to the average degree $\langle k \rangle$)
- The tail decreases exponentially, which indicates that nodes with degree k very different from the average degree are rare
- The clustering coefficient is independent of a node's degree



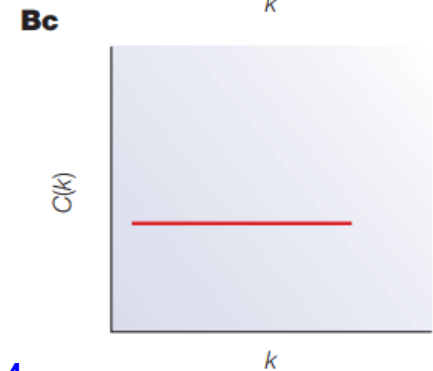
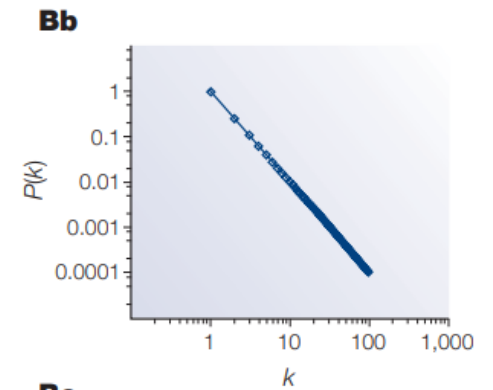
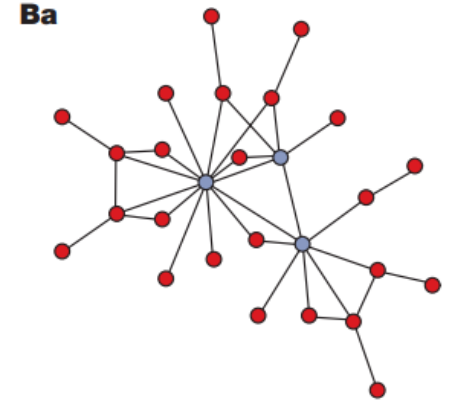
Barabasi et al. Nature Review
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Scale-free networks



- Scale-free networks are characterized by a **power-law degree distribution**
- The probability that a node has k links follows $P(k) \sim k^{-\gamma}$, where γ is the degree exponent (typically $\gamma \in [2, 3]$)
- The probability that a node is highly connected is statistically more significant than in a random graph
- In the **Barabási-Albert model**, at each time point a node with M links is added to the network, which connects to an already existing node i with probability $\Pi_i = k_i / \sum_j k_j$
- The underlying mechanism is that nodes with many links have higher probability of getting more (this is also referred to as **preferential attachment**)

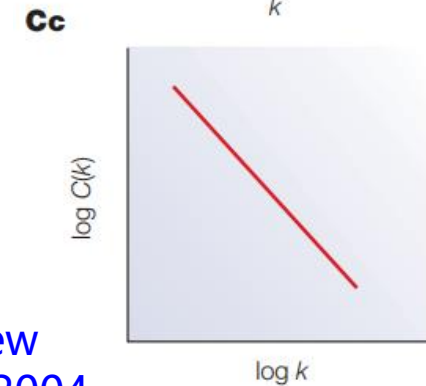
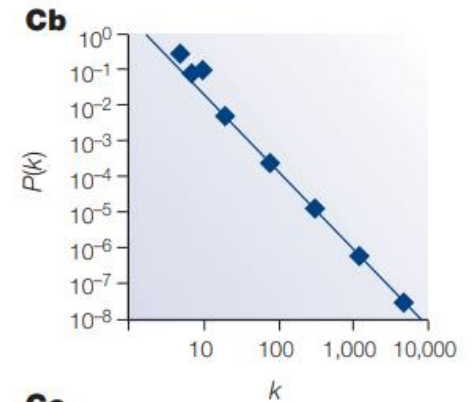
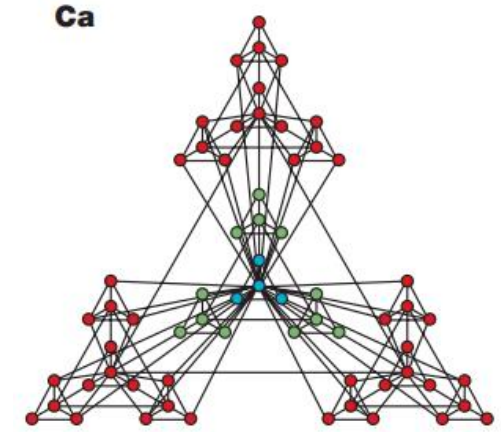


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Hierarchical networks



- A hierarchical structure arises in systems that combine **modularity** and **scale-free topology**
- The hierarchical model in the right-side figure (Ca) is based on the replication of a small cluster of four nodes (the central ones)
- The external nodes of the replicates are linked to the central node of the original cluster
- The resulting network has a **power-law degree distribution**, thus it is **scale-free**
- The average clustering coefficient scales with the degree, following $C(k) \sim k^{-1}$



Barabasi et al. Nature Review
Genetics, 101(5), 101-113, 2004





Topology of biological networks



- An extensive commentary has been published by Albert in 2005, reviewing the literature on the topology of different kinds of biological networks

Réka Albert, Scale free networks in cell biology, *Journal of Cell Science*, 118(21), 4947-4957, 2005

- Experimental evidences are reviewed for metabolic, transcriptional regulatory, signal transduction, and functional association networks
- All of the considered networks approximately exhibit power-law degree distribution, at least for the in-degree or out-degree
- For instance, transcriptional regulation networks
 - Exhibit a **scale-free out-degree distribution**, signifying the potential of transcription factors (TFs) to regulate multiple targets
 - On the other hand, their **in-degree distribution** is a more restricted exponential function, suggesting that combinatorial regulation by several TFs is less frequent





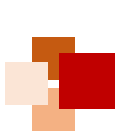
Biological meanings of nodes and edges



- Depending on the kind of biological network, the edges and nodes of the graph have different meanings
- Metabolic network
 - nodes: metabolic product, edge: a reaction transforming A into B
- Transcriptional regulatory network (protein-DNA interaction)
 - nodes: genes and proteins, edge: a TF regulates a gene
- Protein-protein interaction network
 - nodes: proteins, edge: interaction between proteins
- Gene regulatory networks (functional association between genes)
 - nodes: genes, edge: expressions of A and B are correlated

Note: Often transcriptional regulatory network and gene regulatory network (GRN) are used interchangeably.

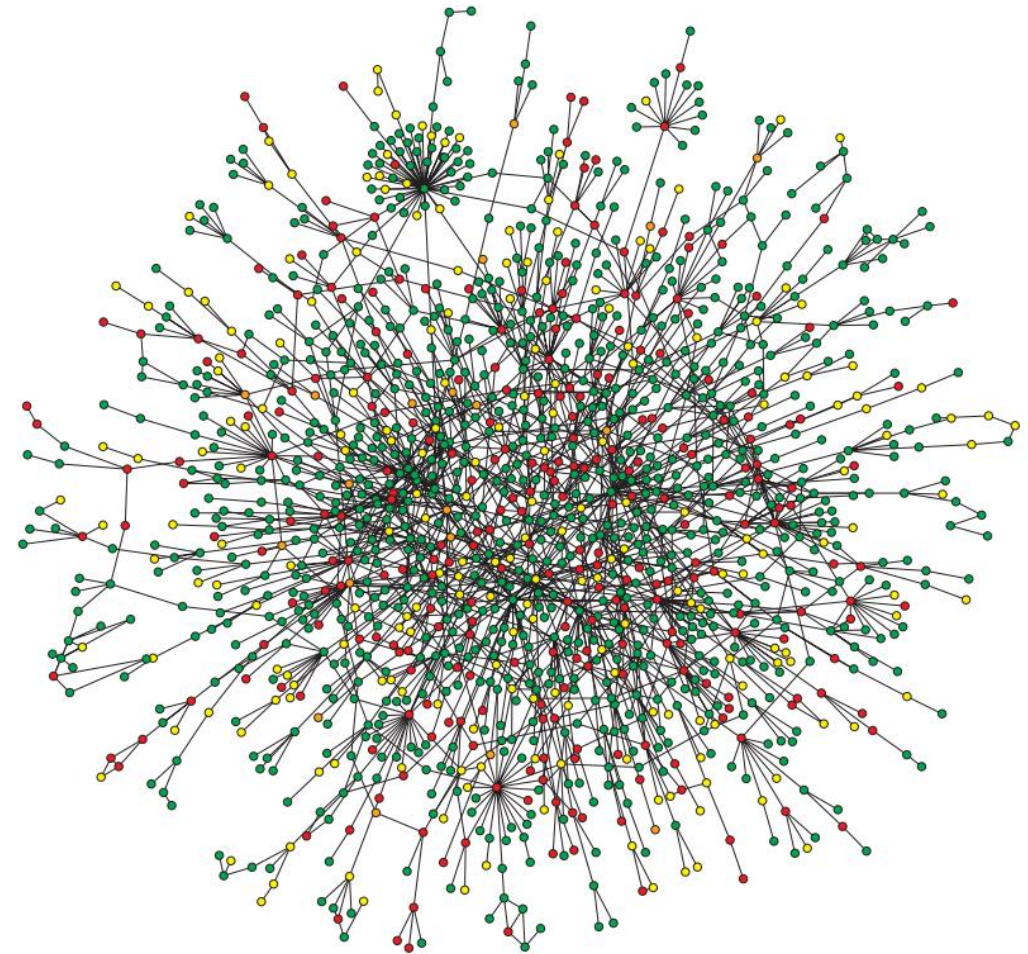




Protein-protein interaction network



- This network is based on **yeast two-hybrid (Y2H) experiments**
- A few highly connected nodes (hubs) hold the network together
- The color of a node indicates the phenotypic effect derived from removing the corresponding protein
 - **Red**: lethal
 - **Green**: non lethal
 - **Orange**: slow growth
 - **Yellow**: unknown



Barabasi et al. Nature Review
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Metabolic reactions



- Metabolic reactions can be divided into two categories:
 - **Catabolic reactions**: breakdown of complex compounds to get energy and building blocks
 - **Anabolic reactions**: assembling of the compounds used by the cellular mechanisms
- Living cells require energy and material for:
 - Building up membranes
 - Storing molecules
 - Replenishing enzymes
 - Replication and repair of DNA
 - Movement





Basic concepts of metabolism



- Historically, metabolism is the part of cell functioning that has been studied more thoroughly than other cellular networks during the past few decades
- This implies that several well-studied mathematical tools exist for describing metabolic networks:
 - **Enzyme kinetics** investigates the dynamic properties of the individual reactions in isolation
 - **Stoichiometric analysis** deals with the balance of compound production and degradation at the network level
 - **Metabolic control analysis** describes the effect of perturbations in the network, in terms of changes of metabolites' connections
- Most of the tools used in the quantitative study of metabolic networks can also be applied to other types of networks





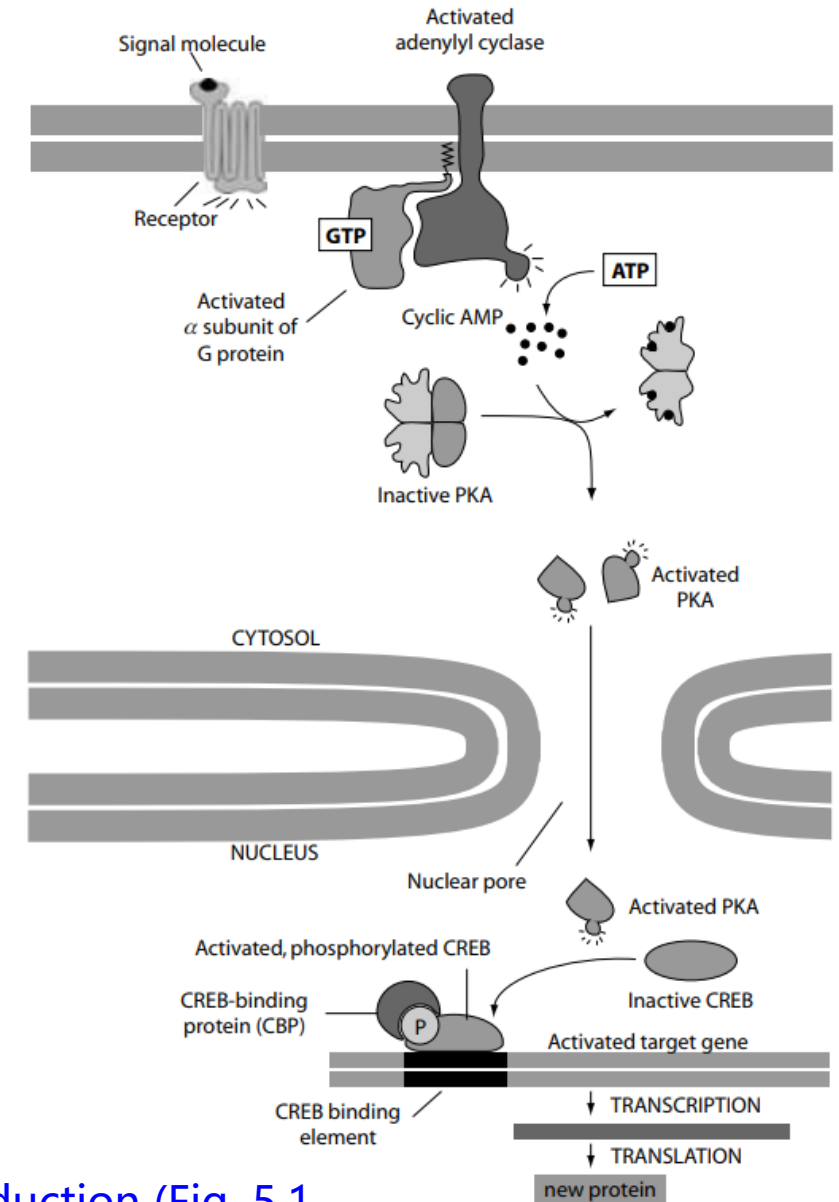
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B. Palsson, *Systems Biology: Properties of Reconstructed Networks*, Cambridge University Press, 2006

Signaling network

- Signaling networks (pathways) involve the transduction of a “signal” from the outside to the inside of a cell
- Three fundamental modes of signal transduction:
 - One cell sends a soluble signal that diffuses to the target cell
 - Cells manipulate the composition of the extracellular matrix
 - Cells communicate through specific and direct cell-to-cell mechanisms
- Comparison with metabolism:
 - Metabolism involves breakdown of molecules for energy production and synthesis of metabolites
 - Signaling conveys “information”, often leading to the change of transcription state of the genome

A 3-level view of signal transduction (Fig. 5.1, Palsson's book 2006)





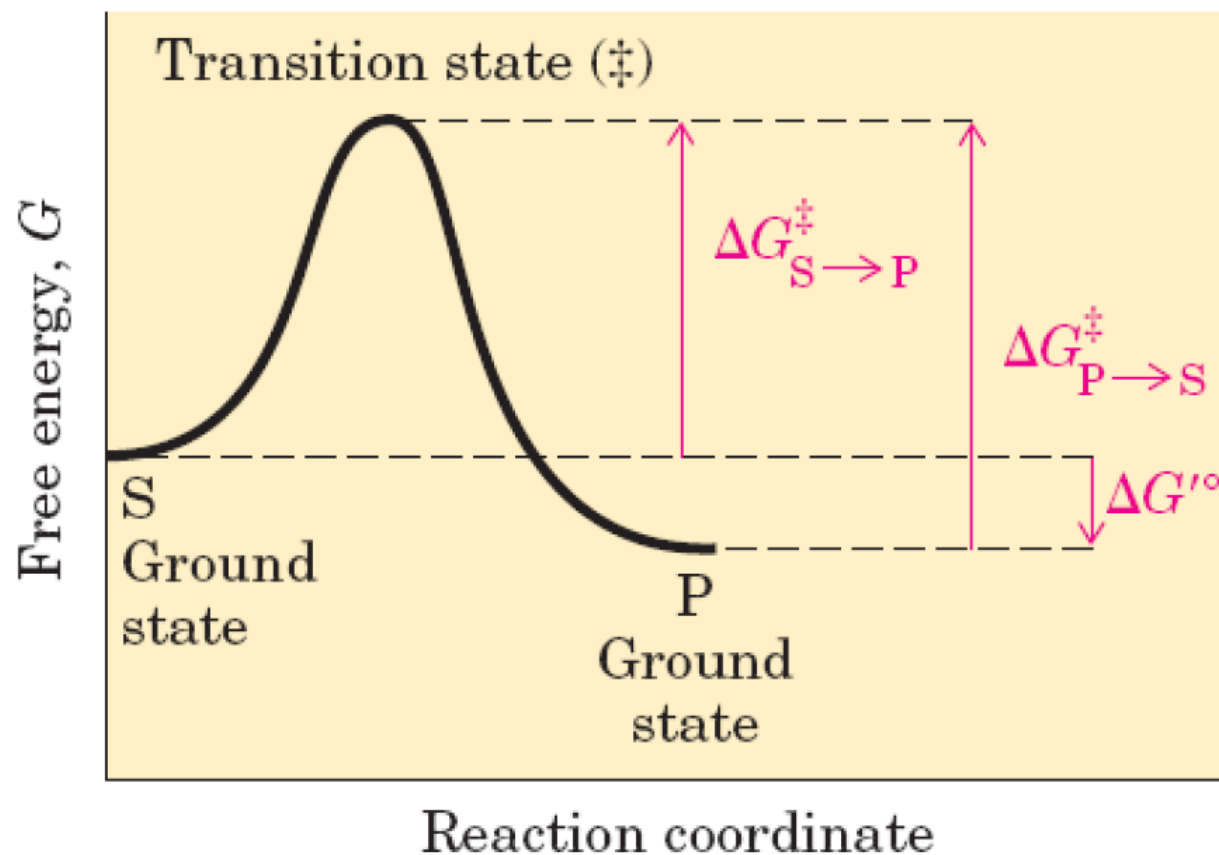
Deterministic Modeling of Biochemical Reactions



Biochemical energy



- The **equilibrium** of a reaction is linked to the variation of biochemical standard **free energy**, $\Delta G'^{\circ}$
- The **velocity**, instead, depends on the **activation energy**, $\Delta G_{S \rightarrow P}^{\ddagger}$





Reaction rate



- The **reaction rate** is determined by
 - The **concentration** of the reactants
 - The **kinetic constant**, usually denoted by k
- The reaction rate for $S \xrightarrow{k} P$ is

$$V = k[S]$$

- From **transition-state theory (TST)**, it is possible to derive the relation

$$k = \frac{kT}{h} e^{-\Delta G^\ddagger/RT}$$

k : Boltzmann constant

h : Planck constant

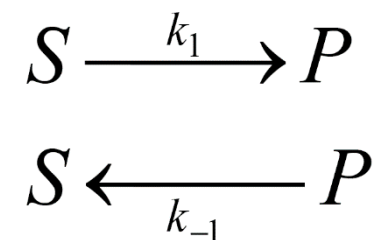


Reaction equilibrium



- In a chemical reaction, **chemical equilibrium** is the state in which both reactants and products are present in concentrations which have no further tendency to change with time

- Let us consider the following basic reversible reaction



- It can be described by the system of ODEs below:

$$\begin{aligned} \frac{d[S]}{dt} &= -k_1[S] + k_{-1}[P] \\ \frac{d[P]}{dt} &= k_1[S] - k_{-1}[P] \end{aligned}$$

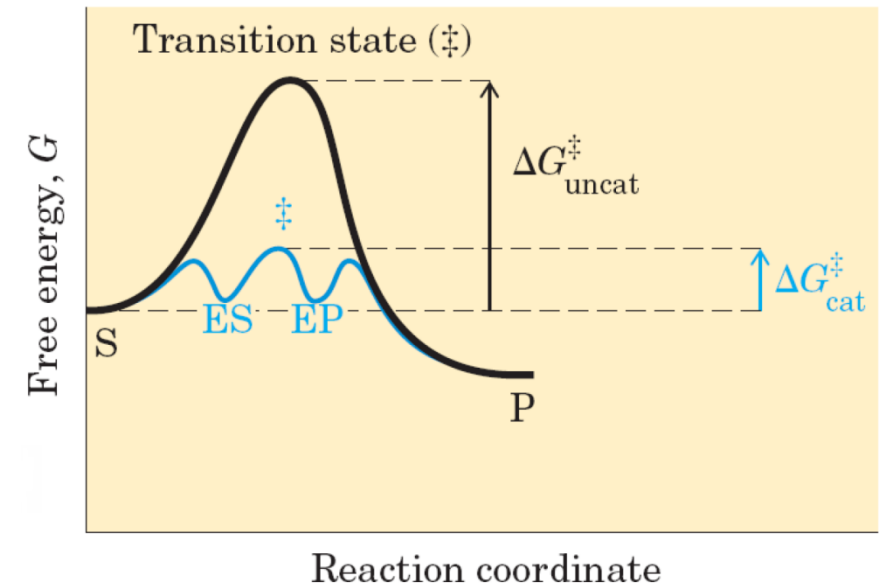


$$K'_{eq} = \frac{[P]}{[S]} = \frac{k_1}{k_{-1}}$$



Enzymatic reaction kinetics

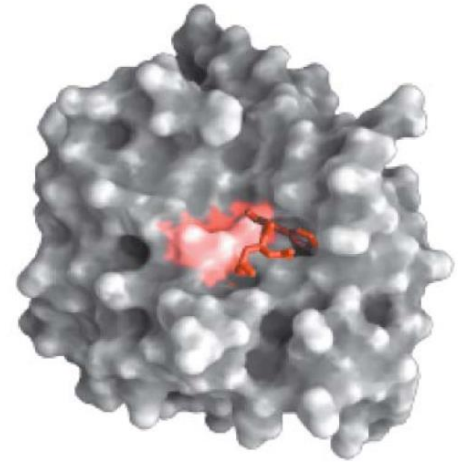
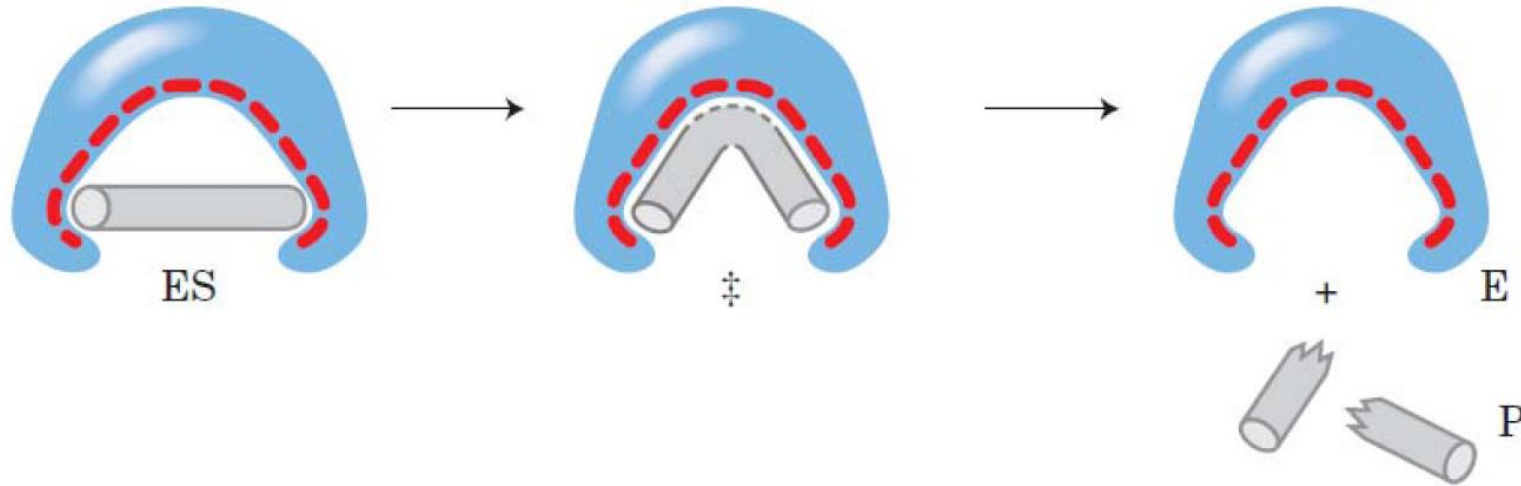
- **Enzymes** are a family of proteins specialized in the catalysis of reactions
- Catalysts do not react and do not affect the reaction equilibrium. However, they increase the reaction rate by decreasing the activation energy
- The basic mechanism is the formation of an enzyme-substrate complex, which creates a more favorable condition for the formation of the product
- After the substrate has been transformed into product, the complex dissociates and the enzyme is free to catalyze the next reaction

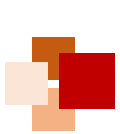


The role of enzymes



- Enzymes play a central role in all the biological processes (e.g. metabolism, regulation, signaling)
- Many diseases are caused by deficiency of some enzyme
- Many drugs act by interacting with enzymes

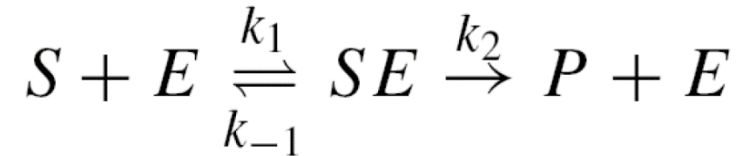




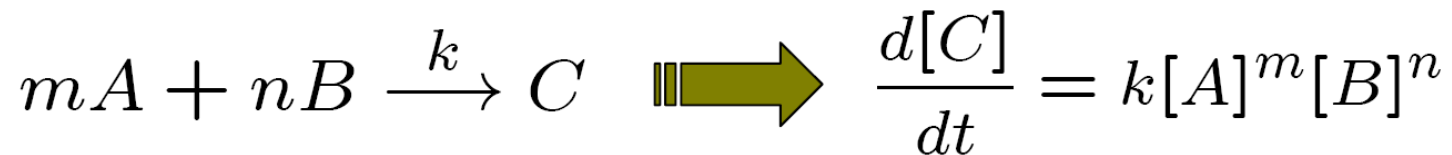
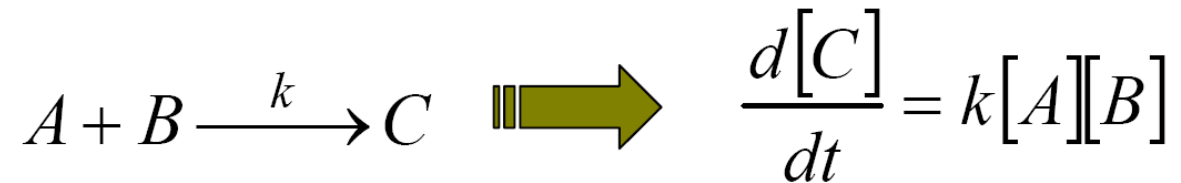
Michaelis-Menten model



- The basic model of enzymatic reaction was proposed by Michaelis and Menten in 1913



- The **law of mass action (LMA)** states that the reaction rate is proportional to the product of the concentrations of the reactants



Leonor Michaelis
(1875 - 1949)



Maud Menten
(1879 - 1960)



Michaelis-Menten model (cont' d)

- Applying the LMA to the model reaction scheme, we obtain

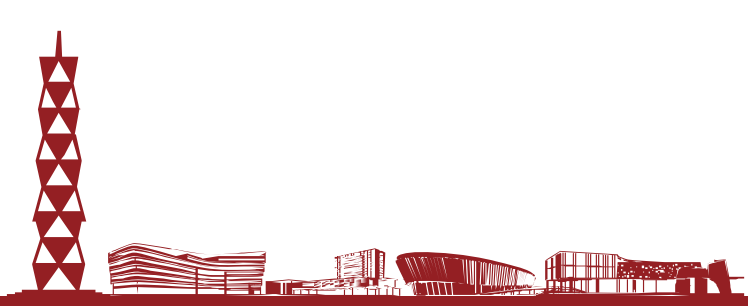
$$\begin{aligned}\dot{s} &= -k_1 es + k_{-1}c, & \dot{e} &= -k_1 es + (k_{-1} + k_2)c \\ \dot{c} &= k_1 es - (k_{-1} + k_2)c, & \dot{p} &= k_2 c\end{aligned}$$

$$s := [S], e := [E], c := [ES], p := [P]$$

with initial conditions: $s(0) = s_0$, $e(0) = e_0$, $c(0) = 0$, $p(0) = 0$

- In the above equations, $\dot{x}$ means the derivative of variable x with respect to time (Newton's notation)
- Note: The equation of $p(t)$ is decoupled and yields

$$p(t) = k_2 \int_0^t c(\tau) d\tau$$



Simplified M-M model



- The total amount of enzyme (free + bound) remains constant over time, i.e. summing the equations of $e(t)$ and $c(t)$ equals a constant

$$\dot{e} + \dot{c} = 0 \quad \Rightarrow \quad e(t) + c(t) = e_0$$

- From the above, we can derive $e(t)$ and substitute into the other equations, obtaining

$$\begin{aligned} \dot{s} &= -k_1 e_0 s + (k_1 s + k_{-1})c, & s(0) &= s_0 \\ \dot{c} &= k_1 e_0 s - (k_1 s + k_{-1} + k_2)c, & c(0) &= 0 \end{aligned}$$



Simplified M-M model (cont' d)

- Typically, the initial formation of the complex ES is much faster than the product formation, hence it can be assumed to be instantaneous
- This assumption lets us pose $dc/dt \cong 0$, that is

$$c(t) = \frac{e_0 s(t)}{s(t) + K_m}$$

where the positive constant

$$K_m = \frac{k_{-1} + k_2}{k_1}$$

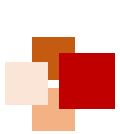
is the **Michaelis-Menten constant**

- Furthermore, the amount of enzyme is typically much less than that of substrate (at least in metabolic reactions), so one can assume that $s(0) = s_0$ also in the simplified model



- In the basic model of enzymatic reaction, we have assumed that each molecule of enzyme only binds one substrate molecule
- However, it is quite common to have multiple binding sites, e.g. the hemoglobin has 4 binding sites for oxygen
- An enzymatic reaction is said to be **cooperative** if the binding of one molecule to one site affects the binding affinity at other sites
- Such reactions are often described with **Hill equations**

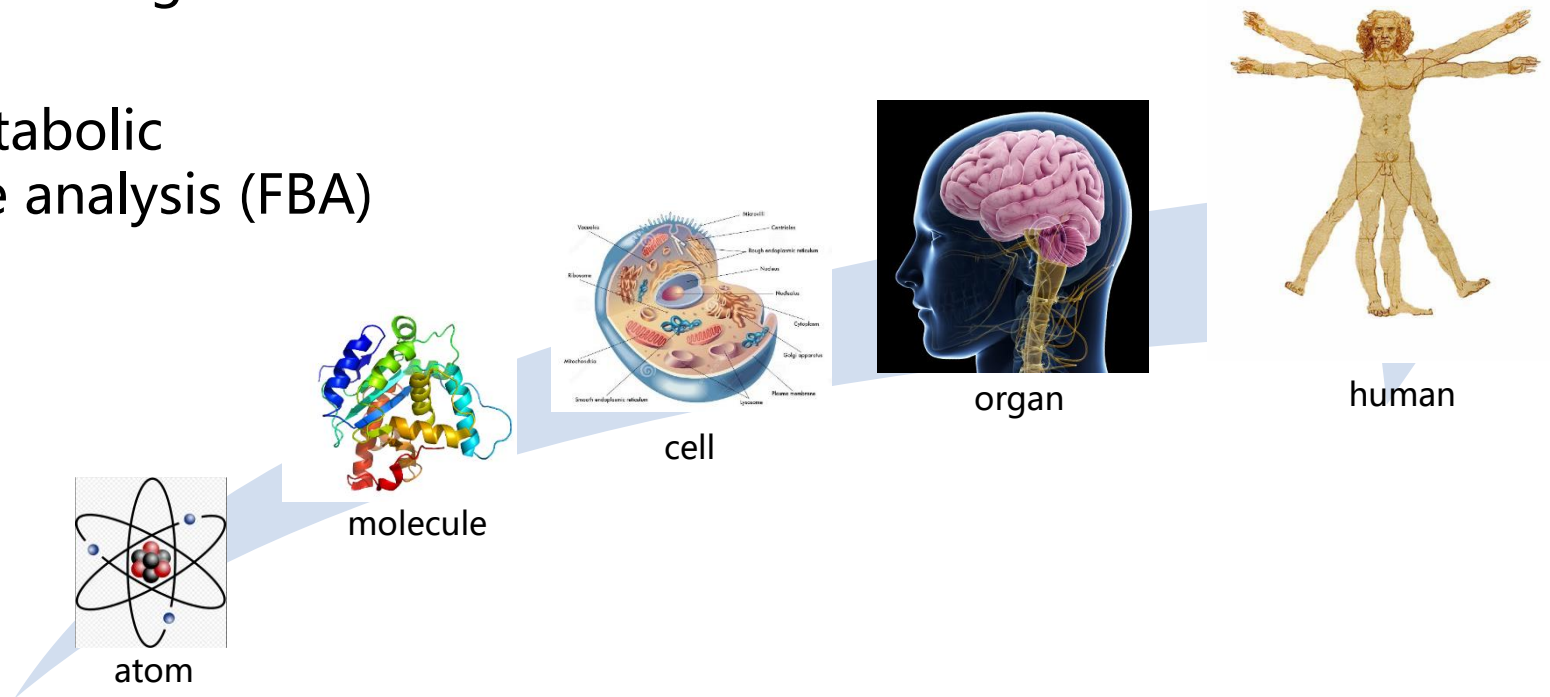




Other topics of systems biology



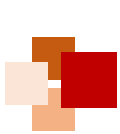
- Integration of multi-modal and multi-omics data (genomics, transcriptomics, proteomics, epigenomics, etc.)
- Inference of biological networks from omics data, e.g. inference of gene regulatory network (GRN) from single-cell gene expression data
- Simulate and predict the metabolic phenotypes, e.g. flux balance analysis (FBA)
- Whole-cell modeling
- Virtual physiological human
- Systems medicine





LLM4GRN: LLM for GRN inference

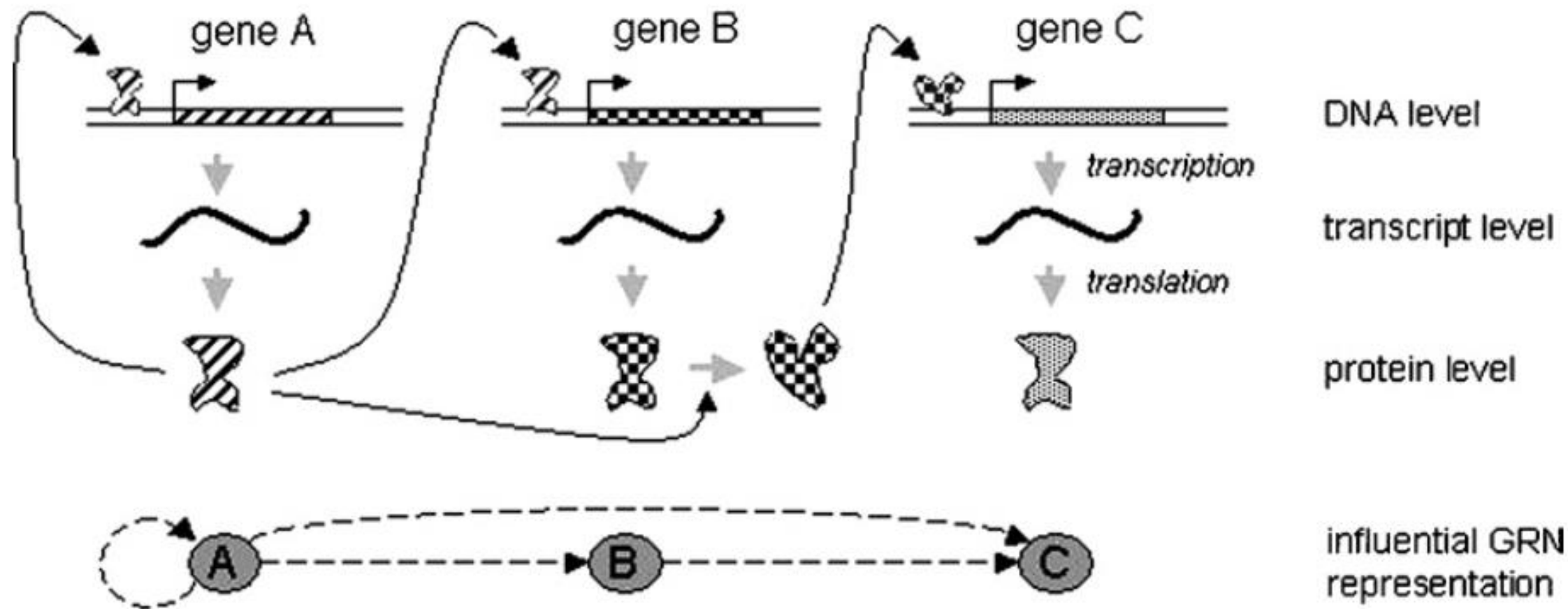




Gene Regulatory Network (GRN)



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Hecker et al. *BioSystems*, 2009

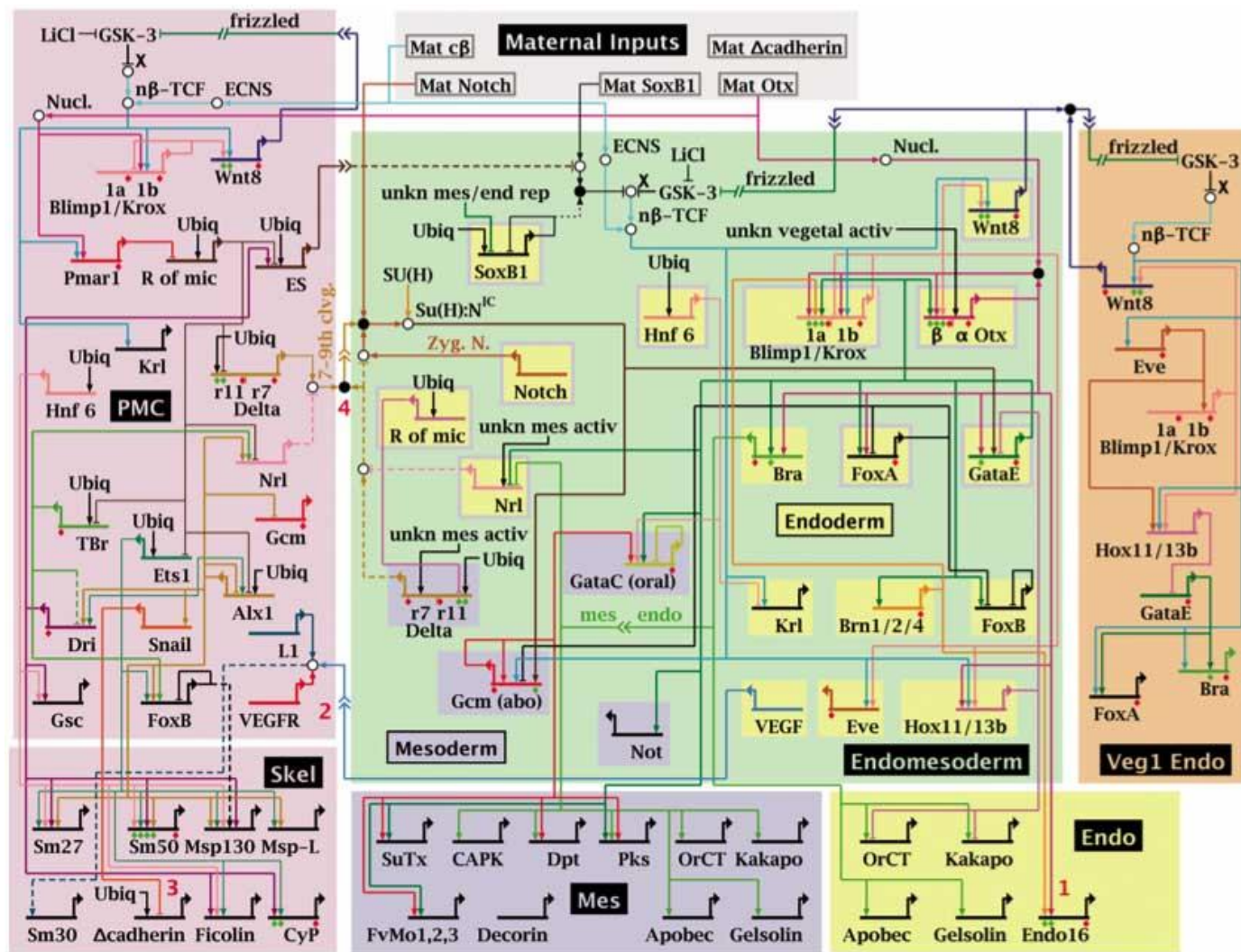


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Biology is becoming a digital science



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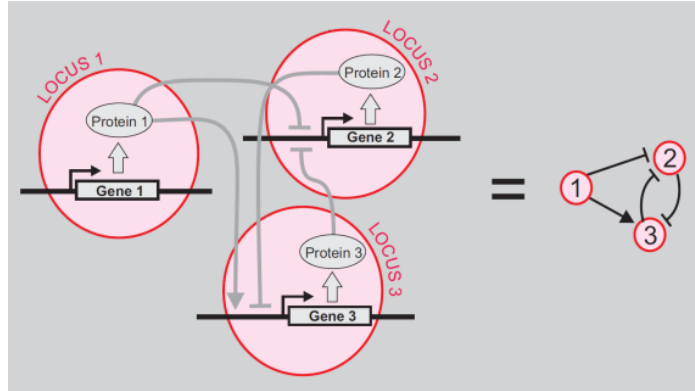
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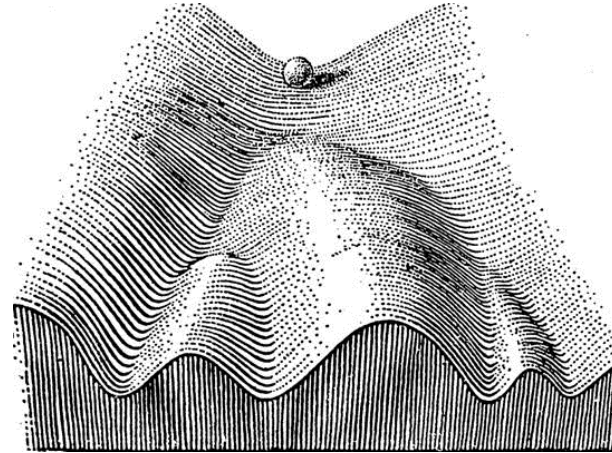
Waddington's epigenetic landscape



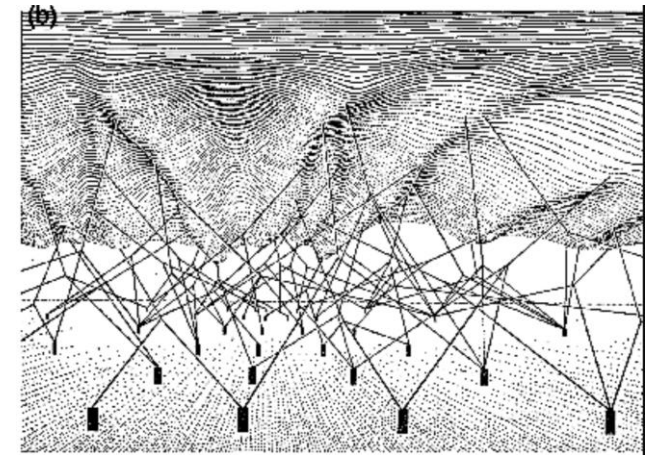
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(A) An example GRN



(B) Waddington's Epigenetic Landscape



The complex system of interactions underlying the epigenetic landscape (Waddington, 1957, Fig. 5, Ref. [2])

Figure: Mapping from genotype to phenotype. (A) Gene regulatory network (GRN) – the circuit controlling the cellular phenotypes (figure from Huang, 2012, Ref. [1]). (B) Waddington's epigenetic landscape – ageing is like a ball rolling down the hill (figure from Waddington, 1957, Ref. [2]).

[1] Huang, S. The molecular and mathematical basis of Waddington's epigenetic landscape: a framework for post-Darwinian biology? *Bioessays* **34**, 149-157, 2012.

[2] Waddington, C. H. The strategy of the genes; a discussion of some aspects of theoretical biology, George Allen & Unwin, 1957.

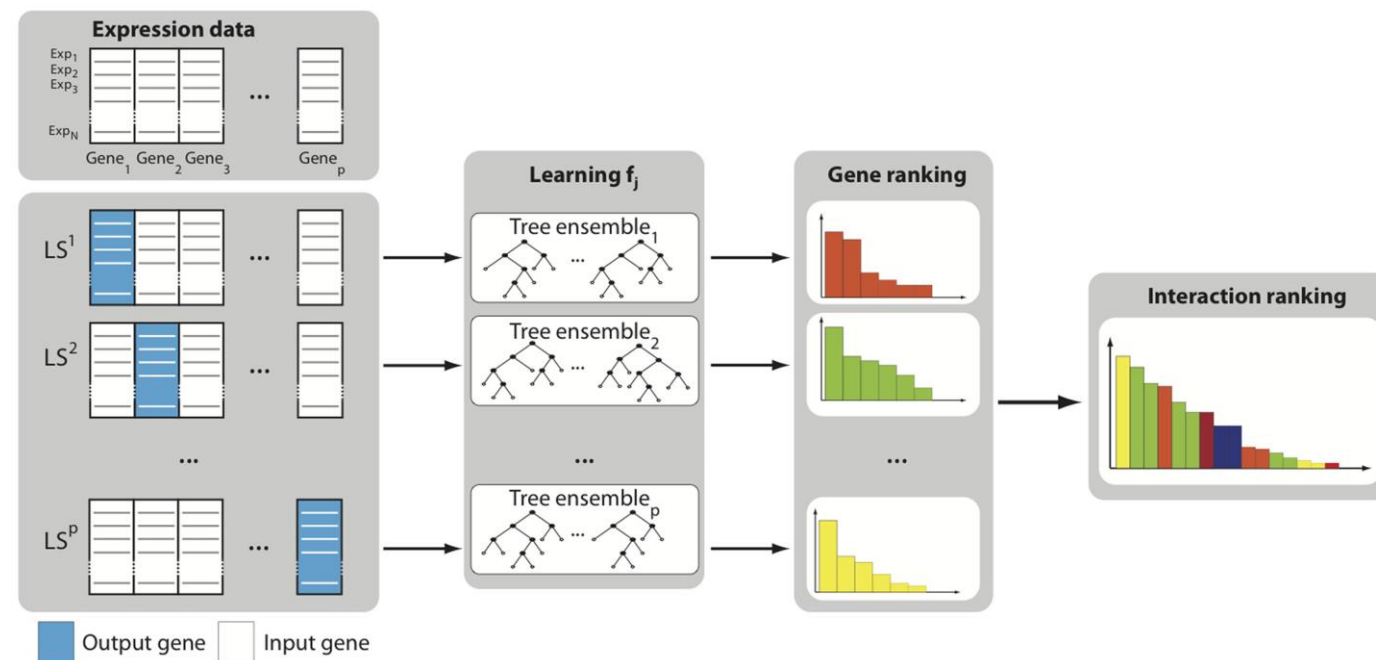


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Algorithms for GRN inference from single-cell transcriptomic data



	Category	Properties			
		Addl. inputs	Time ordered?	Directed?	Signed?
PIDC	MI	—	✗	✗	✗
GENIE3	RF	—	✗	✓	✗
GRNBOOST2	RF	—	✗	✓	✗
SCODE	ODE + Reg	ODE parameters	✓	✓	✓
PPCOR	Corr	—	✗	✗	✓
SINCERITIES	Reg	—	✓	✓	✓
SCRIBE	MI	Type of RDI	✓	✓	✗
SINGE	GC	Regression parameters	✓	✓	✗
LEAP	Corr	Lag	✓	✓	✗
GRISLI	ODE + Reg	Regression parameters	✓	✓	✗
GRNVBEM	Reg	—	✓	✓	✓
SCNS	Bool	Boolean model parameters	✓	✓	✓



GENIE3 procedure

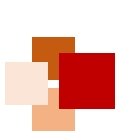
Pratapa, A., Jaliha, A.P., Law, J.N. et al. (2020) Benchmarking algorithms for gene regulatory network inference from single-cell transcriptomic data. Nature Methods 17, 147–154. <https://doi.org/10.1038/s41592-019-0690-6>

Huynh-Thu VA, Irrthum A, Wehenkel L, Geurts P (2010) Inferring Regulatory Networks from Expression Data Using Tree-Based Methods. PLoS ONE 5(9): e12776. doi:[10.1371/journal.pone.0012776](https://doi.org/10.1371/journal.pone.0012776)



- 集成LLMs和知识库进行GRN推理
- 利用LLM捕捉复杂生物相互作用
- 由于缺乏真实的Ground truth, 作者使用因果合成数据生成任务评估模型的性能





按照**知识库**和**GRN推理**方式的**区别**可以分为四种设置：

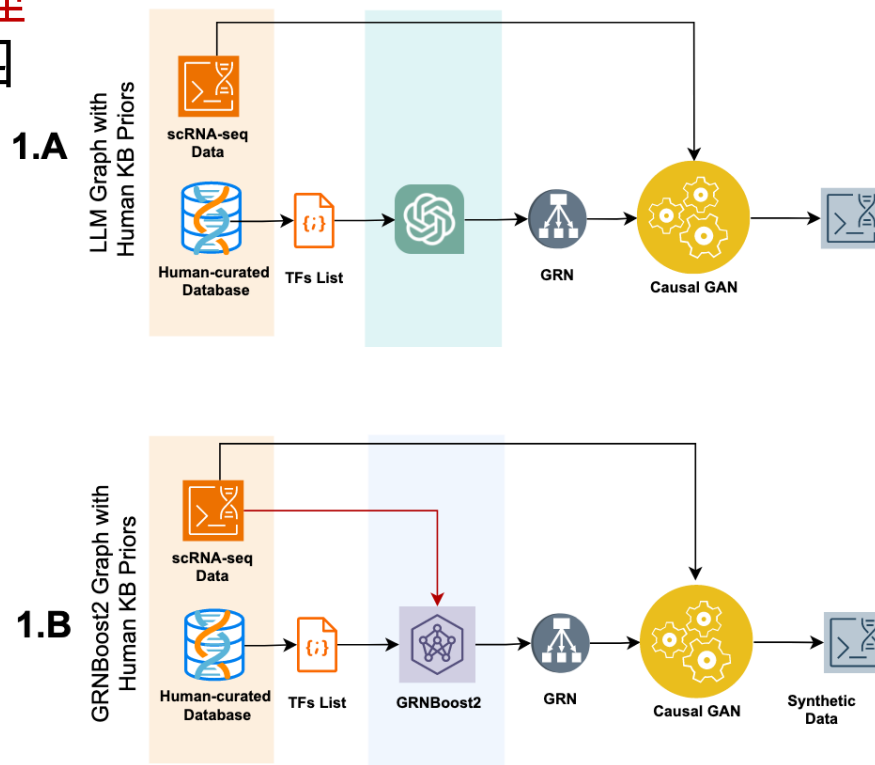
1.A 使用**人类筛选的****知识库**和**LLM**

1.B 将1.A的LLM更换为**GRNBoost2**

2.A 基于**LLM本身的****知识库**和**LLM**

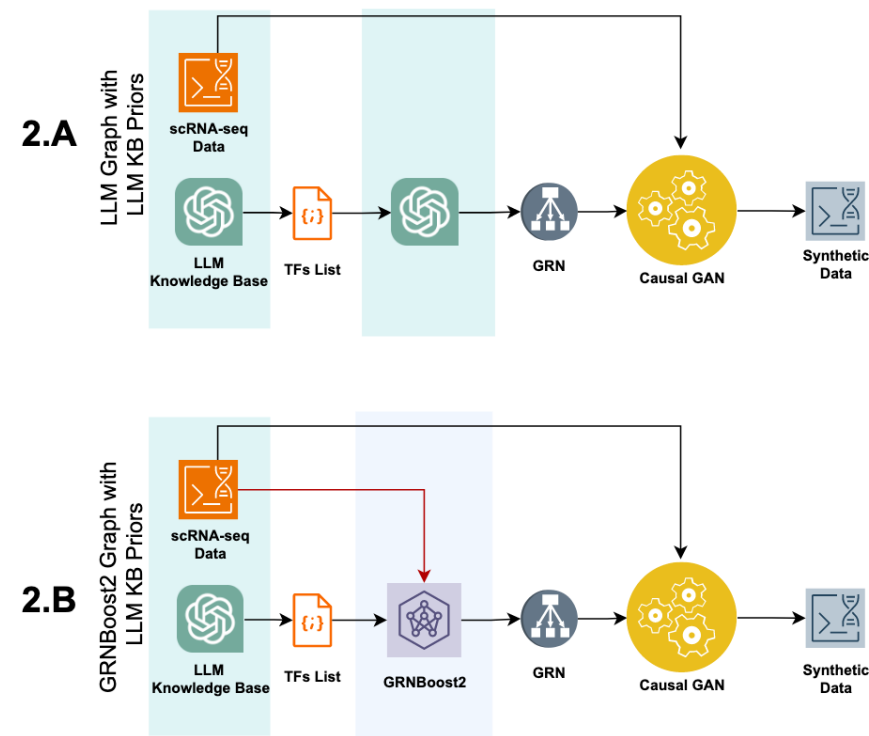
2.B 将2.A的LLM更换为**GRNBoost2**

Setting 1: Human Knowledge Base



(a) Setting 1

Setting 2: LLM Knowledge Base

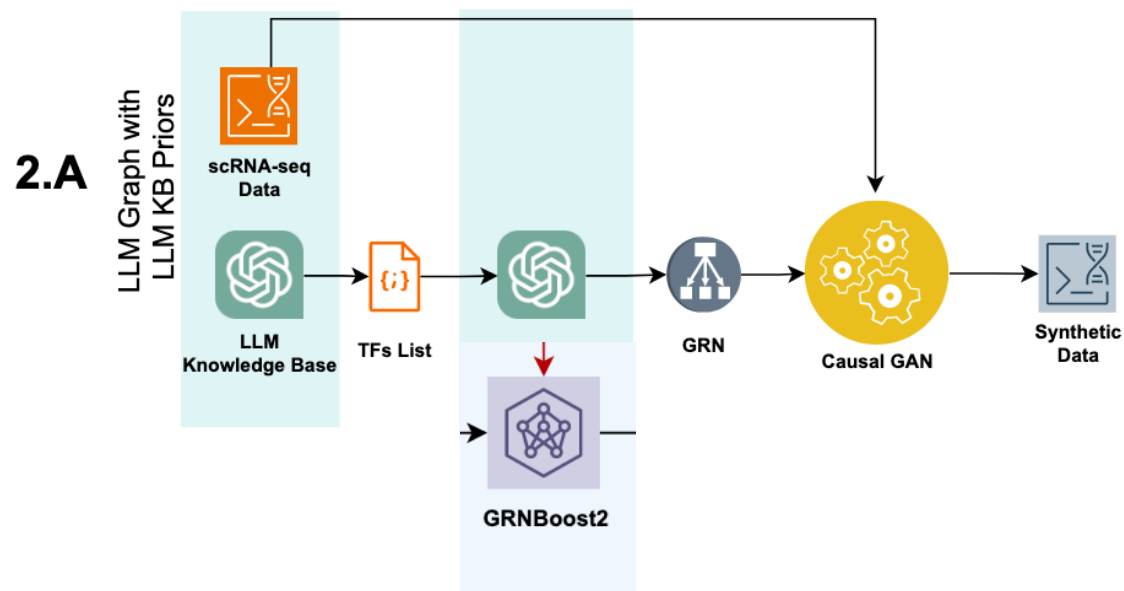
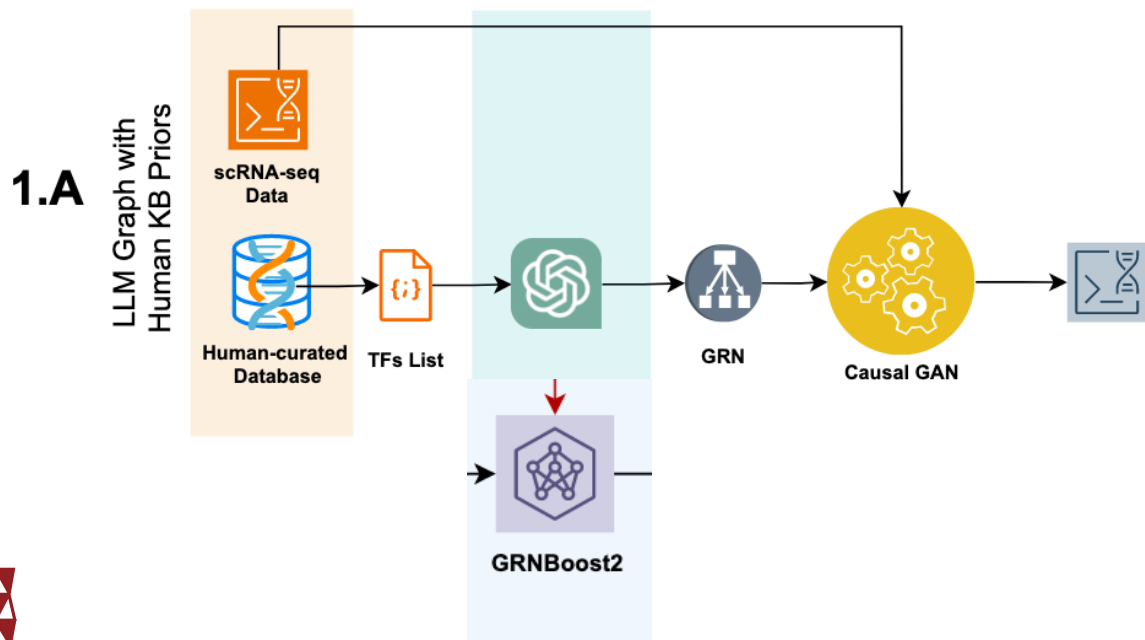


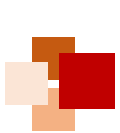
(b) Setting 2

LLM4GRN Pipeline



1. 通过从知识库中确定一组转录因子 T 以及它们所调控的靶基因 R
2. 进而为LLM或GRNBoost2提供 TF 候选列表
3. 将 TF 候选列表输入到 LLM 或 GRNBoost2（包括scRNA-Seq数据）中推理得到GRN
4. 将推理得到的GRN结合scRNA-Seq数据输入到GRouNdGAN获取合成数据





提示词模板 (prompt templates)



Prompt

We need to find transcriptomic factors related to CONTEXT. What are the transcriptomic factors genes related to GENE-X out of LIST-OF-TFs. Which of these 10 TFs have a causal relationship with GENE-X gene? Do not include any genes beyond these. Give potential candidates. Think step by step and return the answer in the format <Answer> [first suggestion, second suggestion, third suggestion, fourth suggestion and so on....] </Answer>. You have to return the 10 potential TFs that are related to the give gene only, otherwise your answer will be disqualified.

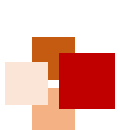
Chain of Thought (CoT) prompting with context provision

Prompt

We need to find transcriptomic factors related to CONTEXT out of given genes. What are the transcriptomic factors genes out of LIST-OF-TFs. Which of these can be transcriptomic factors? Do not include any TFs beyond these. Give potential candidates. Return the answer in the format <Answer> [first suggestion, second suggestion, third suggestion, fourth suggestion and so on....] </Answer>. Describe the reasoning first and then return answer in requested format with the potential TFs.

Extracting potential TFs from the gene list





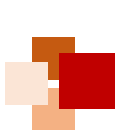
- PBMC-A11: 人类外周血单核细胞 (#68579)
- PBMC-CTL: 来自PBMC-A11的CD8+ Cytotoxic T-cells (#20773)
- BoneMarrow: 小鼠骨髓造血干细胞 (#2730)

Dataset	# TFs	# Targets	# Genes	# Possible Edges	# Imposed Edges	GRN density Edges
PBMC	75	925	1000	69375	9250	0.1333
CTL	65	935	1000	60775	9350	0.1538
BoneMarrow	68	932	1000	63376	9320	0.1471

Table 3: The density of the causal graphs.

在后续的合成数据生成评估任务中，分别从前两个数据集中抽取1000个细胞，在第三个数据集中抽取500个细胞作为对照组





在多个随机种子情况下不同生成的GRN的一致性 (Setting 1, 基于人类筛选的知识库)

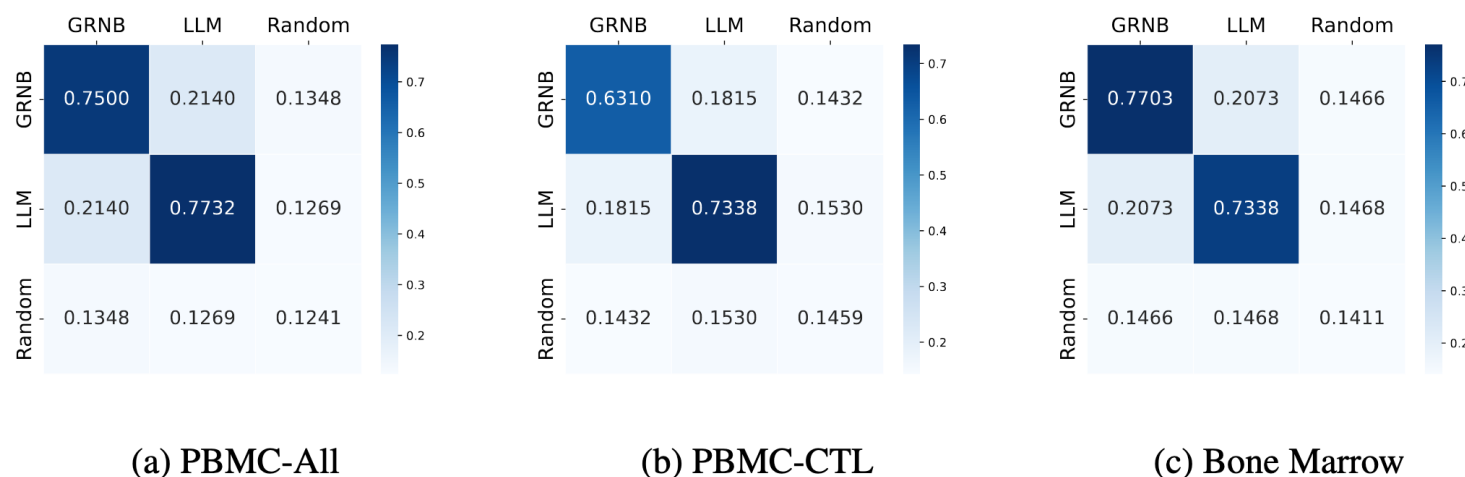


Figure 2: Overlap between GRNs proposed by different methods. LLM demonstrates a higher self-overlap compared to GRNBoost2 algorithm.

(1) 与GRNBoost2 (GRNB) 生成的GRN相比, LLM生成的GRN表现出更高的鲁棒性, 这体现出LLM具有相对更高的确定性。

(2) GPT-4生成的GRN和随机GRN之间的重叠小于GPT-4生成的GRN和GRNBoost2生成的GRN之间的重叠, 表明GPT-4可以生成有意义的图。





■ 评估指标

1. 欧几里得距离: $d(r, s) = \|\mu(R) - \mu(S)\|_2$ $\mu(R) = \left(\frac{1}{n} \sum_{i=1}^n R_{i,j} \right)_{j=1, \dots, m}$ and $\mu(S) = \left(\frac{1}{n} \sum_{i=1}^n S_{i,j} \right)_{j=1, \dots, m}$
2. 余弦相似度: $d_{\cos}(r, s) = 1 - \frac{\mu(R) \cdot \mu(S)}{\|\mu(R)\|_2 \|\mu(S)\|_2}$
3. 最大平均距离: 是一种非参数双样本检验, 用于确定样本是否来自同一分布。
4. RF-AUROC: 用于评估模型区分真实数据集和合成数据集的能力。采用随机森林 (RF) 分类器, 并使用接收器工作特性 (AUROC) 曲线下的面积来评估真实和模拟细胞是否可以有效区分。





下游合成数据生成任务评估



Setting 1

		Cosine distance ↓	Euclidean distance ↓	MMD ↓	RF AUROC ↓
PBMC-All	Setting 1 KB ^H				
	GPT-4	<u>0.00024</u> ±0.00004	<u>89</u> ±7	<u>0.0072</u> ±0.001	<u>0.63</u> ±0.043
	GRNBoost2	0.00047±0.00021	121±25	0.0139±0.006	0.73±0.050
	Random	0.00045±0.00013	121±22	0.0166±0.003	0.85±0.019
PBMC-CTL	Setting 1 KB ^H				
	GPT-4	0.00265±0.00253	176±115	0.0238±0.020	0.79±0.206
	GRNBoost2	<u>0.00025</u> ±0.00004	<u>65</u> ±6	<u>0.0053</u> ±0.000	0.59 ±0.028
	Random	0.00027±0.0001	66±7	0.0085±0.001	0.75±0.021
BoneMarrow	Setting 1 KB ^H				
	GPT-4	0.00238±0.00052	86±9	0.0124±0.001	0.70±0.080
	GRNBoost2	0.00190 ±0.00023	77 ±5	0.0118 ±0.001	0.64 ±0.023
	Random	0.00219±0.00030	82±6	0.0150±0.003	0.75±0.041

在Setting 1的场景下，GPT-4推断的GRN在PBMC-All的所有指标中都达到了最佳性能。





下游合成数据生成任务评估



Setting 2

		Cosine distance ↓	Euclidean distance ↓	MMD ↓	RF AUROC ↓
BoneMarrow PBMC-CTL PBMC-All	<i>Setting 2 KB^{GPT4}</i>				
	GPT-4	0.00026±0.00009	90±13	0.0206±0.001	0.86±0.018
	GRNBoost2	0.00023 ±0.00008	83 ±17	0.0069 ±0.001	0.59 ±0.028
	Random	0.00026±0.00011	92±14	0.0226±0.002	0.87±0.018
	<i>Setting 2 KB^{GPT4}</i>				
	GPT-4	0.00028±0.00009	67±11	0.0067±0.001	0.70±0.025
	GRNBoost2	0.00020 ±0.00004	57 ±6	0.0049 ±0.000	0.59 ±0.019
	Random	0.00022±0.00005	59±7	0.0080±0.000	0.76±0.023
	<i>Setting 2 KB^{GPT4}</i>				
	GPT-4	0.00217±0.00050	82±10	0.0137±0.003	0.72±0.044
	GRNBoost2	<u>0.00193</u> ±0.00023	<u>78</u> ±4	<u>0.0119</u> ±0.001	0.64 ±0.033
	Random	0.00297±0.00080	96±13	0.0172±0.004	0.79±0.046

在Setting 2场景下，GRNBoost2推断的GRN在PBMC数据集上的所有指标中都达到了最佳性能，另外，BoneMarrow数据集上，GRNBoost2在不同KB下的表现都更好。

这些发现表明，将LLM衍生的先验与GRNBoost2的统计推断相结合，可以更准确、更稳健地表示基因调控网络（GRN）。





Recap



- In this lecture, we have learned:
 - Basic principles and challenges of systems biology (e.g. construct predictive models, data integration)
 - Different types of biological networks (metabolic network, signaling network, protein-protein interaction network, gene regulatory network, etc.)
 - Topological characteristics (e.g. degree distribution, clustering coefficient) and structures of biological networks (e.g. scale-free, hierarchical)
 - Basics of deterministic modeling, e.g. Michaelis-Menten model
 - Basics of stochastic modeling, e.g. the Gillespie algorithm





References



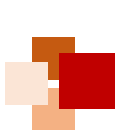
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- B. Palsson (2006) "Systems Biology: Properties of Reconstructed Networks" , Cambridge University Press.
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- Daniel T. Gillespie (1976) "A general method for numerically simulating the stochastic time evolution of coupled chemical reactions" , *Journal of Computational Physics*, 22: 403–434. (**The Gillespie Algorithm**)





Appendix: Stochastic Modeling

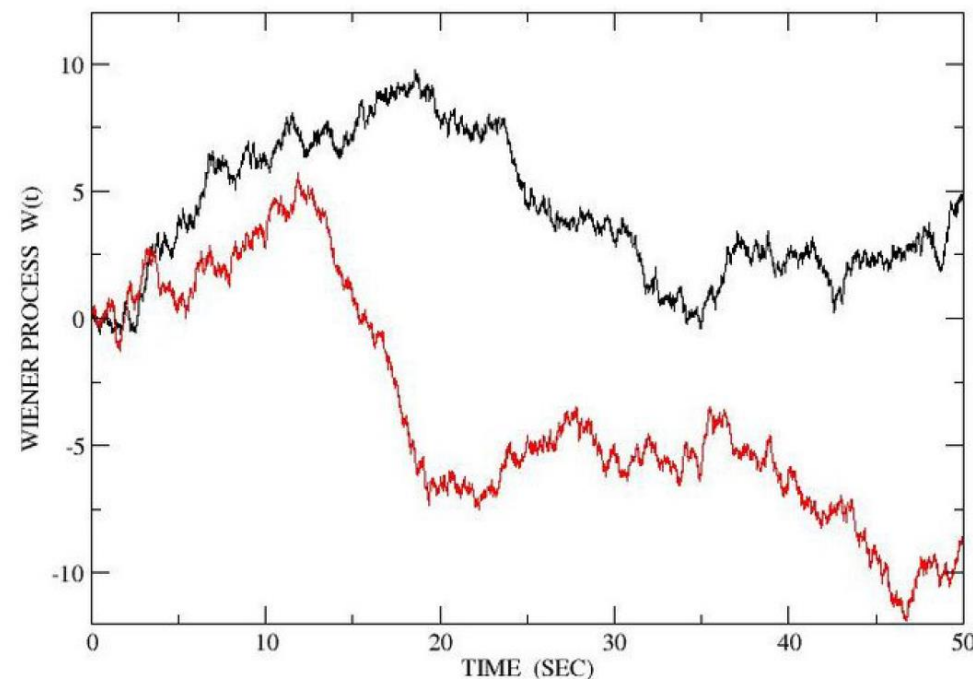




Stochastic models



- The models considered so far are **deterministic**, i.e. given the initial conditions (the state at t_0) and the exogenous signals perturbing the model, the response is univocally determined
- A **stochastic** model, on the contrary, involves **random variables**, therefore its behavior cannot be predicted *a priori*, although it can be statistically characterized
- The figure shows two realizations of the same stochastic process, starting from the same initial condition





Biological systems are stochastic



- Certainly biological systems fall in the category of stochastic systems, because the very basic steps of every molecular reaction can be described **only** in terms of its **probability of occurrence**
- Moreover, the diffusion of molecules is a realization of a random walk process (**Brownian motion**)
- So, why are deterministic models still so widespread?
 - Usually, the phenomena under consideration involve **a large number of molecules**, therefore the average effect is well described through deterministic equations
- When is it necessary to use stochastic models/simulations?
 - When the mechanism to be described is based on the interaction of **few molecules**, or we want to simulate the functioning of a little pool of cells





Stochastic vs. deterministic constants



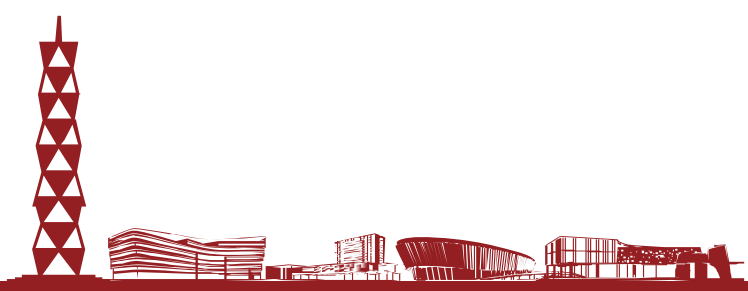
- Intuitively, the **stochastic reaction constant**, c_μ , must be linked to the **kinetic constant** of the corresponding deterministic equation, k_μ
- In the validity ranges of the deterministic models, it is possible to establish the simple relation

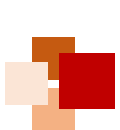
$$k_1 = Vc_1$$

- The presence of the factor V (volume) depends on the fact that deterministic models take as state variables the **concentrations**, whereas stochastic ones use **the number of molecules**



- There are two main approaches to solving the stochastic system and compute the time evolution of the system
 - Master Equation Approach
 - Stochastic Simulation Algorithm (SSA) with Gillespie' s method





Master equation approach



- The key element in this approach is the probability function

$$P(X_1, X_2, \dots, X_N; t)$$

Probability that at time t
there are X_1 molecules of
species S_1 , ..., X_N of species S_N

- The master equation describes the evolution of the function P
- The probability $P(X_1, \dots, X_N; t + dt)$ can be derived as a combination of the probability of all the possible reactions that can happen within dt
- The differential equation as derived by this argument usually does not have an analytical solution, nor does it admit a computationally efficient solution



Gillespie algorithm



- The algorithm was first presented in
 - D. T. Gillespie (1976). A general method for numerically simulating the stochastic time evolution of coupled chemical reactions.
Journal of Computational Physics, 22: 403-434.
- The purpose is to simulate chemical reactions with limited computational resources
- The outcome is a realization of the underlying stochastic process





Basic principles of Gillespie algorithm



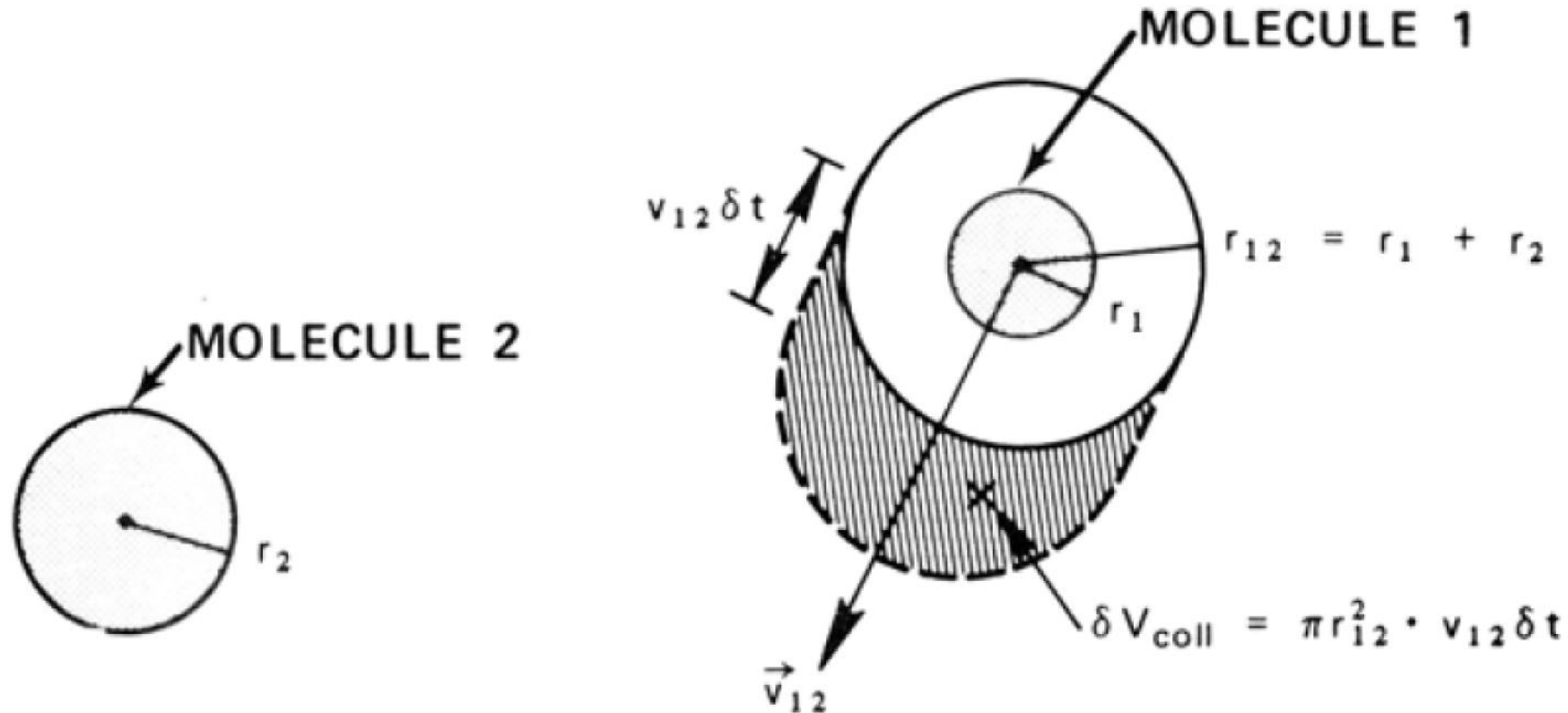
- The basic working principles of the Gillespie algorithm are derived from the description of the collision of particles in a vessel
- When two molecules collide, the reaction happens only if they have proper orientation and kinetic energy
- The first hypothesis is that the frequency of non-reacting collision is much larger than that of reacting ones
- Secondly, the algorithm assumes that each reaction involves no more than two molecules



■ Sketch of the derivation



- The method starts from the observation of the volume occupied by **molecule 1** that is in relative motion with respect to another one, **molecule 2**





Sketch of the derivation (cont' d)



- Assuming a random uniform distribution of the molecules in the volume V , the **probability of collision** can be computed as

$$\delta V_{\text{coll}}/V = V^{-1} \pi r_{12}^2 \bar{v}_{12} \delta t$$

Probability of collision of two molecules in the time interval δt

- If in such a volume we have X_1 molecules of the species S_1 and X_2 of the species S_2 , the probability of their collision is

$$X_1 X_2 V^{-1} \pi r_{12}^2 \bar{v}_{12} \delta t$$





Steps of the Gillespie algorithm



- **Step 0 (Initialization)**: Define the number of molecules of each species, the kinetic constants and the random number generator
- **Step 1 (Monte Carlo)**: Generate random numbers, to determine which is the next reaction and the length of the time interval dt
- **Step 2 (Update)**: Increase time by dt and update the number of molecules of each species on the basis of the occurred reaction
- **Step 3 (Iterate)**: If the number of reactants is greater than zero and the simulation stop time has not yet been reached, iterate from Step 1

